Treasury PCA Butterflies, Carry, and Relative-Value Discipline

Dhruv Kohli

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Abstract

This paper asks whether a PCA-neutral 2s5s10s Treasury butterfly isolates a tradable curvature residual or only creates a clean-looking factor signal. I estimate Treasury principal components on daily yield changes, construct exact and regularized butterflies, test stationarity and walk-forward stability, and then add carry/rolldown, transaction costs, volatility targeting, and an actual-bond implementation check. The evidence is intentionally sobering. The exact full-sample butterfly is neutral to level and slope by construction, but the spread is not strongly stationary, rolling exact hedge ratios become unstable, and a stricter mean-reversion strategy loses money after realistic filters. A carry-aligned curvature-momentum rule is more researchable: it earns 307.47 bp with a 0.18 annualized Sharpe and a smaller drawdown, but it is still not production alpha. The paper's main contribution is therefore the research workflow. It shows how to move from a factor residual to a trade candidate without confusing factor neutrality with edge.

1 Introduction

Treasury relative-value trades often start from a simple intuition: broad yield-curve moves can be decomposed into level, slope, and curvature, and a butterfly can isolate the residual curvature component. In desk language, a 2s5s10s butterfly trades the belly against the wings. The trade is appealing because it appears to remove the most obvious macro duration bets while keeping exposure to curve shape.

The empirical question is whether that residual is actually tradable. A butterfly can be neutral to the first two principal components and still fail as a strategy. The residual may not be stationary, hedge ratios may depend on the sample, carry may work against mean reversion, transaction costs may dominate, and fixed-maturity yields may not map cleanly to actual Treasury securities. This paper makes those frictions explicit.

The project has two layers. The first layer reconstructs the classical PCA butterfly carefully. It uses yield changes rather than yield levels, compares correlation PCA with covariance PCA, solves 2s5s10s weights that neutralize level and slope, and tests the resulting spread for stationarity and mean reversion. The second layer turns the factor residual into a trading-research object. It re-estimates hedge ratios walk-forward, introduces a regularized gross-capped hedge, adds carry/rolldown and cost-aware sizing, and checks whether nearest-bond implementation preserves the fixed-maturity signal.

The main finding is not that the naive butterfly is a finished trade. The cleaned-up evidence rejects that story. The ADF test does not strongly reject a unit root in the static spread, the estimated half-life is about 1,222 trading days, exact rolling PCA weights can become explosive, and the enhanced mean-reversion specification loses 372.69 bp after costs. A carry-aligned curvature-momentum rule is more plausible because it respects the spread's observed persistence and the curve's own rolldown, but even that rule has only a modest Sharpe. The right conclusion is to continue the research around regularized hedge construction, carry-aware signals, and actual bond execution, not to present a naive z-score as alpha.

The remainder of the paper is organized as follows. Section 2 places the design in the fixed-income factor and relative-value literatures. Section 3 describes the data. Section 4 defines the PCA, butterfly, and strategy objects. Section 5 reports the empirical results. Section 6 discusses the implementation gap. Section 7 concludes, and Appendix A gives a claim-by-claim audit of what the repo proves.

2 Related Literature

The factor structure follows Litterman and Scheinkman (1991), who show that level, slope, and curvature explain most Treasury yield-curve variation. This paper uses that result as a starting point rather than as a trading conclusion. A factor-neutral portfolio removes exposure to the estimated factors; it does not guarantee stationarity, positive carry, or executable P&L.

The return-predictability motivation is related to the fixed-income literature on bond risk premia and yield-curve factors, including Cochrane and Piazzesi (2005). The trading-friction layer is closer to fixed-income arbitrage work such as Duarte, Longstaff, and Yu (2007), where apparent relative-value opportunities must be interpreted through leverage, drawdown, funding, and implementation constraints. The empirical design also borrows the basic discipline of out-of-sample testing: hedge ratios and z-scores must be lagged or re-estimated walk-forward rather than fit once with future information.

3 Data

The fixed-maturity yield panel is the Gurkaynak–Sack–Wright Treasury curve. The local file contains 16,097 observations from 1961-06-14 through 2025-12-26. The actual-bond panel contains 204,868 bond-date observations from 2022-01-03 through 2025-12-31. Raw workbooks are not tracked in Git. The public repository contains the executed notebook, reusable Python functions, tests, documentation, figures, and this paper.

The core workbooks are `gsw_yields.xlsx` and `treasury_panel_pca.xlsx`. The first file supports fixed-maturity PCA and strategy research. The second supports the implementation check that maps target maturities to nearby Treasury securities. Optional public FRED data can be downloaded by the repo script for credit, stress, and curve-regime context, but downloaded CSV files remain under `data/raw/` and outside version control.

4 Empirical Framework

4.1 PCA on Yield Changes

Let Δy_t denote the vector of daily Treasury yield changes across maturities. The baseline PCA is

$$\Delta y_t = L f_t + \epsilon_t, \tag{1}$$

where the columns of L are factor loadings and f_t are daily factor shocks. The notebook compares two variants. Correlation PCA standardizes each maturity and is useful for interpretation. Covariance PCA keeps yield-basis-point units and is more natural for hedge construction because one basis point at each maturity remains one basis point.

The paper uses yield changes instead of yield levels. Yield levels are highly persistent and can make PCA describe slow-moving curve location rather than tradable risk shocks. The purpose of the factor model is to hedge daily curve-risk exposure, so the shock object is the appropriate input.

4.2 Exact Butterfly Construction

The exact 2s5s10s butterfly uses covariance PCA loadings. It solves for weights

$$w = (w_2, w_5, w_{10})$$

such that

$$w^\top L_1 = 0, \qquad w^\top L_2 = 0, \qquad w_5 = -1. \tag{2}$$

The final condition normalizes the trade as short the 5Y belly. The spread is

$$s_t = w_2 y_{2,t} + w_5 y_{5,t} + w_{10} y_{10,t}. \tag{3}$$

The trading signal is a lagged rolling z-score,

$$z_t = \frac{s_t - \mu_{t-1}^{(126)}}{\sigma_{t-1}^{(126)}}. \tag{4}$$

The lag matters because a z-score that uses today's close to form today's position leaks information from the P&L interval into the signal.

4.3 Regularized Hedge Construction

Exact rolling neutrality can be fragile. When the factor loadings shift, the solution that sets PC1 and PC2 exposure exactly to zero can require large offsetting wing weights. The upgraded specification therefore solves a regularized problem that keeps the 5Y belly weight fixed at -1, penalizes residual level and slope exposure, and penalizes gross notional. In words, the strategy accepts small residual factor exposure in exchange for more stable hedge ratios.

This tradeoff is not a modeling detail. It is a trading decision. A hedge that is exactly neutral but requires unstable gross exposure can be worse than a hedge that is slightly imperfect but implementable and robust.

4.4 Strategy Hypotheses

The paper tests two strategy hypotheses. The first is mean reversion: if the factor residual is rich or cheap, fade it after robust z-score, carry/rolldown, volatility-targeting, and cost filters. The second is carry-aligned momentum: if curvature has been trending and rolldown points in the same direction, follow the move rather than fight it.

This hierarchy is important. A relative-value project should not assume that residuals mean-revert because a chart looks stationary. It should test the residual, reject the mean-reversion thesis if the data do not support it, and then move to a better economic hypothesis.

5 Results

5.1 Treasury Factor Structure

Both PCA variants recover the standard Treasury factor structure. The first three components explain almost all daily yield-change variation:

Factor	Correlation PCA variance	Covariance PCA variance
PC1	88.09%	88.12%
PC2	8.67%	8.69%
PC3	2.32%	2.28%

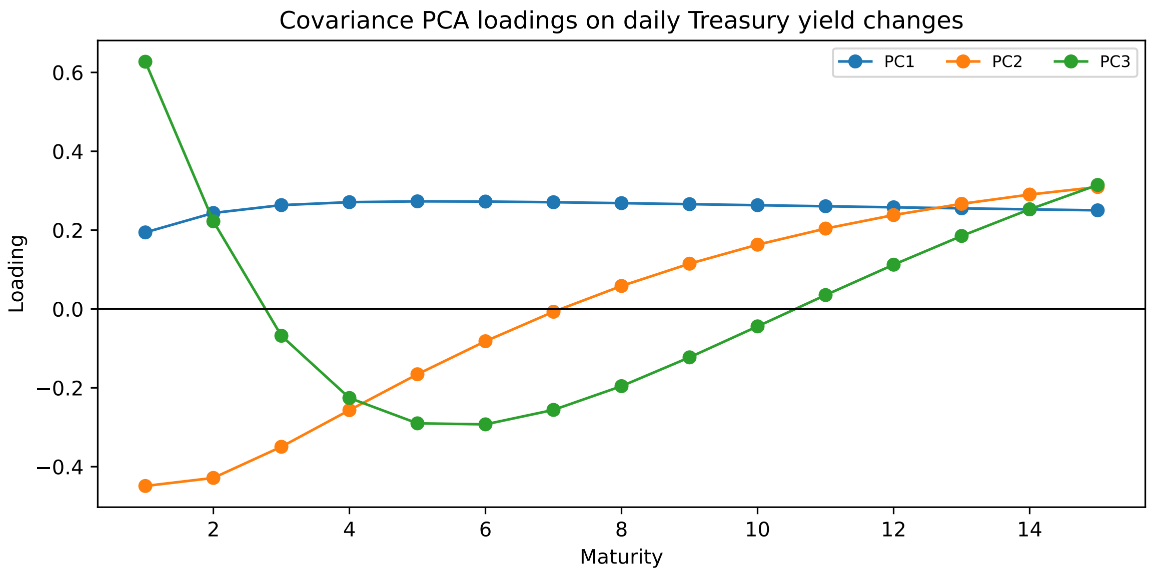

Figure 1: Covariance PCA loadings estimated from daily Treasury yield changes. The first three components line up with level, slope, and curvature, which motivates the 2s5s10s butterfly construction.

This result validates the factor setup. It does not validate the trade. The next step is to ask whether the residual spread has properties a trader can use.

5.2 Exact Butterfly and Stationarity

The full-sample exact PCA-neutral weights are:

Maturity	Exact weight
2Y	0.5777
5Y	-1.0000
10Y	0.5026

These weights neutralize PC1 and PC2 up to numerical precision while retaining PC3 exposure of 0.3963. The construction therefore works mechanically: it produces a curvature residual.

The stationarity evidence is weaker. The ADF statistic is -2.316 with a p-value of 0.1669, and the estimated half-life is about 1,222 trading days. A static lagged-z-score backtest earns 303 bp total with annualized Sharpe 0.20, max drawdown -158 bp, 10,736 active days, and 617 trades. The signal is not useless, but the evidence is not strong enough to support an aggressive mean-reversion claim.

5.3 Walk-Forward Instability

A full-sample PCA backtest uses future information. The notebook therefore re-estimates PCA weights monthly using only the prior three years of yield changes. This repair exposes a different problem: exact rolling weights can become unstable. The exact rolling-weight distribution is:

Maturity	Mean weight	Std	Min	Max
2Y	0.8066	1.2355	-8.5137	17.2066
5Y	-1.0000	0.0000	-1.0000	-1.0000
10Y	0.3578	0.9859	-12.9131	7.7876

The wide range is itself an empirical result. It shows that local PCA loadings can shift enough to make exact neutrality expensive. The walk-forward exact backtest earns 989 bp total P&L, but the annualized Sharpe is only 0.023 and max drawdown is -3,852 bp. Total P&L is therefore the wrong headline. The headline is instability.

5.4 Regularized Hedge Weights

The regularized full-sample hedge is less pure but more stable:

Maturity	Regularized weight
2Y	0.4047
5Y	-1.0000
10Y	0.2407

Gross exposure falls from 2.08 to 1.65. Residual factor exposures are no longer exactly zero (PC1 -0.1111, PC2 0.0314, PC3 0.3694), but the hedge becomes more implementable. A monthly walk-forward version has average weights of 0.3993, -1.0000, and 0.2363 for the 2Y, 5Y, and 10Y legs. Gross exposure averages 1.64 and never exceeds 1.86 in the tested sample.

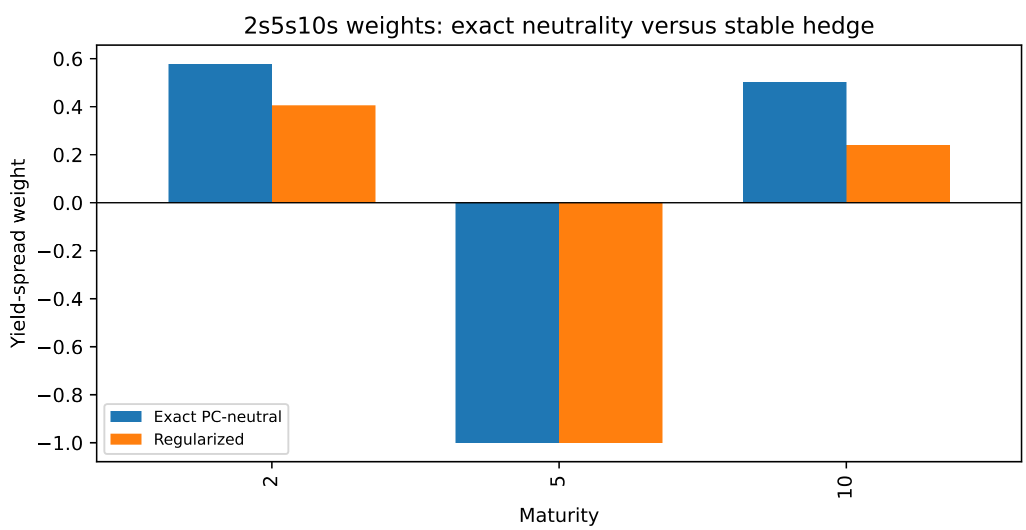

Figure 2: Exact PC-neutral weights versus regularized weights. The regularized hedge sacrifices exact neutrality to reduce gross exposure and stabilize the trade for walk-forward use.

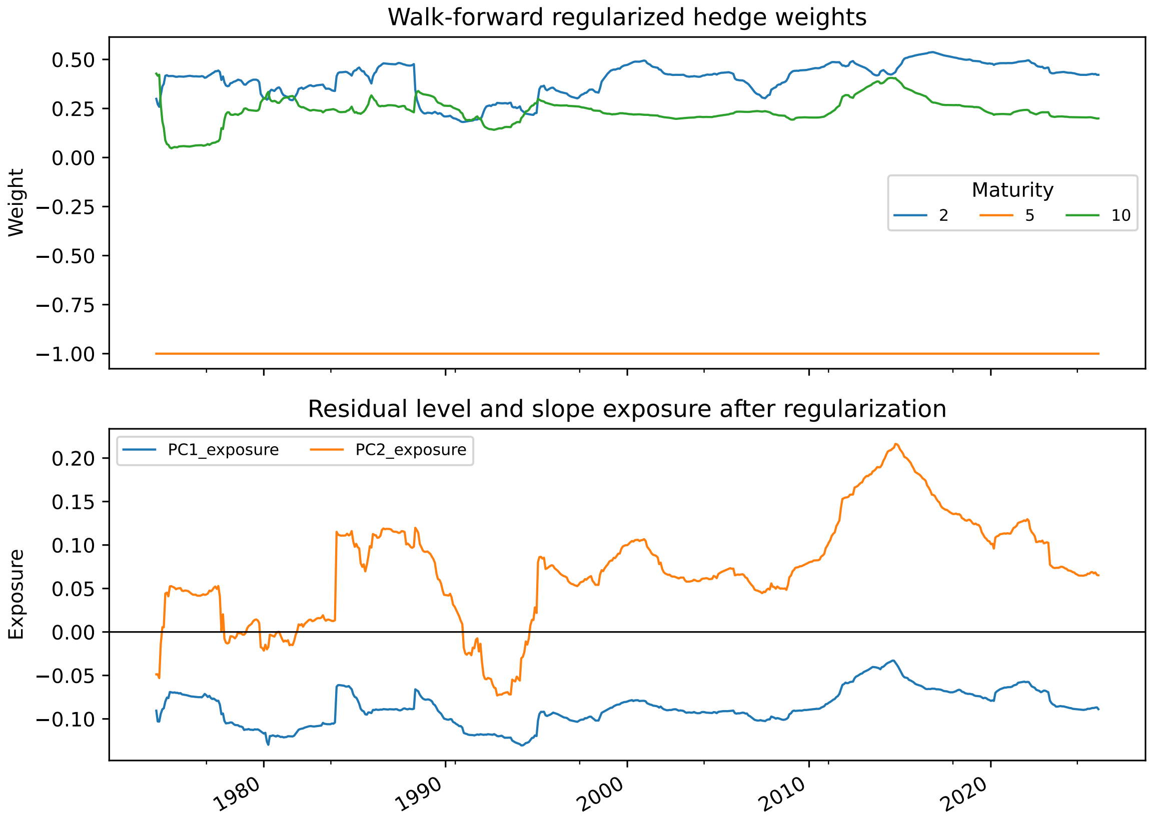

Figure 3: Walk-forward regularized weights and residual PC exposures. The regularized construction prevents the explosive wing weights that appeared in the exact rolling PCA hedge.

5.5 Strategy Comparison

The enhanced mean-reversion strategy adds robust z-scores, carry/rolldown gating, volatility-targeted sizing, and transaction costs. This stricter version loses money:

Strategy hypothesis	Total P&L bp	Annualized Sharpe	Max drawdown bp
Regularized mean reversion	-372.69	-0.36	-454.19
Rolling regularized mean reversion	-449.28	-0.50	-493.25
63-day carry-aligned momentum	307.47	0.18	-125.14

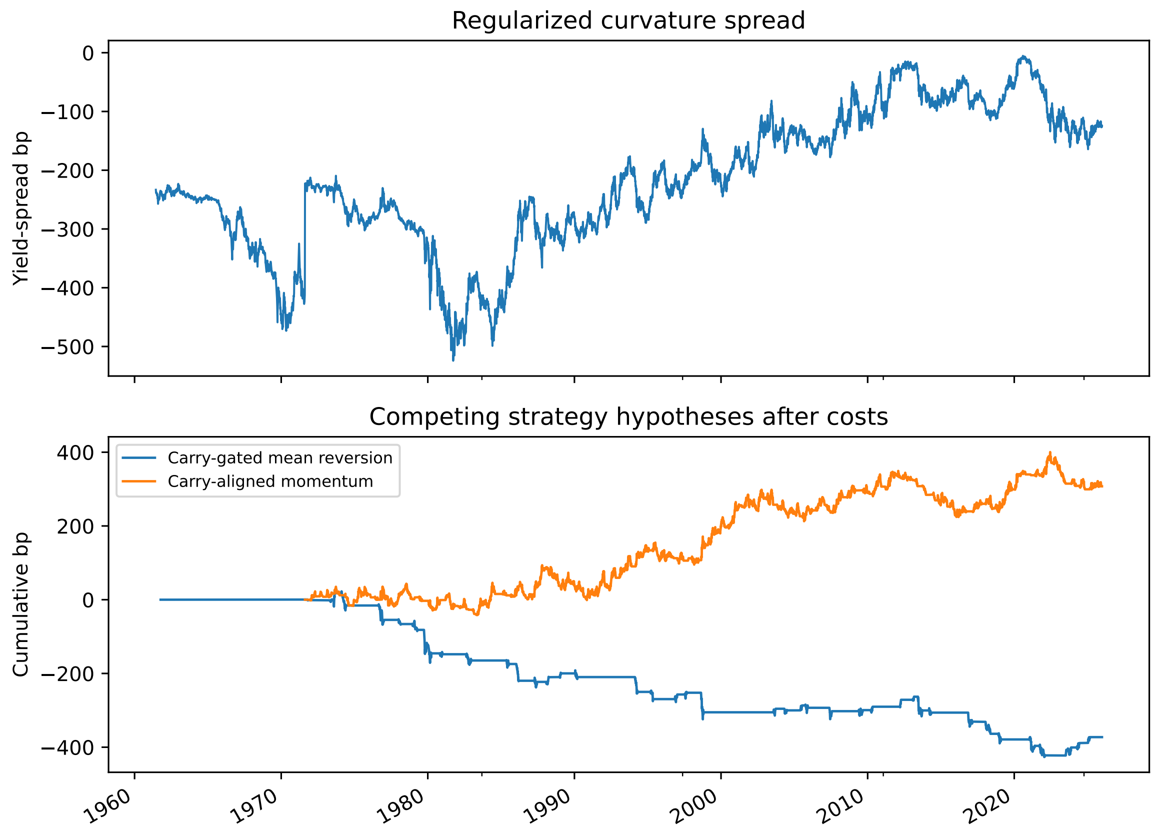

Figure 4: Regularized curvature spread and competing strategy hypotheses. Mean reversion weakens after costs and carry gating, while carry-aligned momentum is the more plausible next research branch.

The table changes the story. It would be easy to present the exact PCA butterfly as a mean-reverting residual and stop there. The stricter evidence says the opposite: factor neutrality is not alpha, and a residual that looks clean may still lose after implementation-aware filters. The more researchable branch is carry-aware curvature momentum, because it aligns the signal with the curve's own rolldown rather than mechanically fading persistence.

6 Implementation Gap

Fixed-maturity yields are clean research objects. Trades are done in securities, futures, swaps, or packages. The notebook therefore maps monthly rebalances to nearby Treasury bonds around the 2Y, 5Y, and 10Y targets and compares bond yield changes against GSW fixed-maturity changes. This section is included because it is where many relative-value projects become unrealistic. Duration, bond selection, accrued interest, repo, financing, bid/ask, roll, and liquidity all matter.

The current repo is production-grade as a research artifact, not as a live trading system. It shows the full path from factor construction to strategy rejection and hypothesis refinement. A production version would need futures or cash-bond execution, duration and DV01 sizing, financing costs, roll mechanics, CTD or benchmark selection, and a portfolio-level risk model. Those are not cosmetic additions. They decide whether the fixed-maturity research signal survives contact with the instrument layer.

7 Conclusion

This paper builds a PCA-neutral Treasury butterfly and asks whether the curvature residual is a tradable relative-value signal. The evidence supports a disciplined answer. The factor model works: Treasury yield changes are dominated by level, slope, and curvature, and exact 2s5s10s weights can neutralize the first two components. The trade thesis is weaker: the static spread is not strongly stationary, exact rolling hedge ratios are unstable, and the enhanced mean-reversion strategy loses money after costs.

The project becomes more interesting after that rejection. Regularized hedge construction produces stable weights, and carry-aligned momentum is a more plausible research branch than naive mean reversion. That conclusion is more valuable for recruiting than an overstated Sharpe. It shows that the research process is allowed to reject the first idea, preserve the useful machinery, and define the next empirical path.

A Claim-by-Claim Methodology Audit

Claim	Implemented evidence	Allowed interpretation
PCA recovers Treasury curve factors	Correlation and covariance PCA on daily yield changes; first three factors explain about 99% of variation.	The factor framework is appropriate for curve-risk decomposition.
Exact butterfly isolates curvature	2s5s10s weights neutralize PC1 and PC2 while retaining PC3 exposure.	The construction creates a curvature residual; it does not prove alpha.
Naive mean reversion is weak	ADF p-value 0.1669, half-life about 1,222 days, and enhanced mean-reversion backtests lose after costs.	The cleaned-up evidence rejects an aggressive z-score mean-reversion story.
Regularization improves implementation quality	Gross exposure falls from 2.08 to 1.65 and rolling regularized gross exposure stays below 1.86.	Small residual factor exposure is a reasonable price for more stable hedge ratios.
Carry-aware momentum is the next branch	63-day carry-aligned momentum earns 307.47 bp with max drawdown -125.14 bp.	The result is researchable but not production alpha.
The public repo is safe to share	Raw workbooks and downloaded public data are ignored; tests and data documentation state the contract.	Recruiters can review methodology and code without private data or secrets being published.

B How to Read the Repository

Start with this paper for the research object and claim hierarchy. Then read `Treasury_PCA_Butterfly_Relative_Value.ipynb` for the executed analysis. The reusable implementation is in `src/treasury_pca_butterfly.py`; tests are in `tests/test_treasury_pca_butterfly.py`; the data contract is in `data/README.md`; and the local knowledge-base protocol is in `docs/knowledge_base_protocol.md`.

References

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